Iron ore into China: delivered cost and the import window

Nathan Couturier. Research note, data to 15 Oct 2026 (daily) and Jul 2026 (monthly benchmark). Interactive version, data and code: https://nathancouturier.github.io/iron-ore-china-arbitrage/

Question

When does it pay to bring a cargo of seaborne iron ore into China, from where, and who captures the margin between the mine and the blast furnace? The note builds the delivered cost of a 62% Fe cargo step by step (FOB netback, Capesize freight, CFR Qingdao, USD/CNY, import VAT, port charges and voyage financing) and compares it with the Dalian futures price for the matching delivery, then asks whether Chinese steel mill margins support the ore price.

Answer

- On 15 Oct 2026 a 62% Fe cargo from West Australia landed in Qingdao at 733 RMB per dry tonne, VAT included, against 692 for 45-day Dalian delivery: a margin of −41 RMB/dmt, so the window is closed.
- The window has almost never been open. On the daily series (from December 2015) the margin exceeded the 25 RMB/dmt cost of using the exchange on 0.9% of trading days and never for more than 6 days in a row; on the monthly baseline it was below the band in 97% of 151 months, by 80 RMB/dmt on average (9.7% of landed cost).
- Gaps fade slowly: the half-life is 5.0 months on monthly data (95% interval 2.3 to 8.2), but a unit root cannot be rejected there (ADF p = 0.16); on daily data the half-life is 23 trading days (ADF p = 0.010). The margin reverts to a negative level, not to zero.
- Freight takes 8.9% of the CFR price from West Australia and 21.2% from Brazil on average since 2014; Brazil's index-equivalent netback trails Australia's by 11.9 USD/dmt on average and by 21.4 in Jul 2026.
- The physical market tracks import parity; the discount sits in Dalian. On 5 dated Fastmarkets Qingdao port prices, Dalian's 45-day price was 72 to 374 RMB/dmt below the port price of 62% fines. Mills are squeezed: the rebar futures margin was −68 RMB/t on 16 Oct 2026, negative on 100% of trading days in 2026, and the pre-registered test finds no sign that low mill margins precede weaker ore.

Data and conventions

Seaborne price: IMF 62% Fe CFR China monthly average (FRED PIORECRUSDM) for the monthly baseline from January 2014; SGX iron ore futures interpolated to a 45-day horizon for the daily series from December 2015 (the current-month contract is not used because it settles on the month's average). SGX contracts from January 2026 settle on Platts IODEX at 61% Fe and are lifted to 62% by the measured monthly spread (mean 2.45 USD/dmt). Dalian: every iron ore contract from the exchange's annual files (2013 to 2024) and Sina (2025 onward), expressed on a 62% Fe basis (standard grade 61% from I2205, plus 10 x X with X = 1.5 RMB per 0.1% Fe in every period), interpolated to a constant 45-day maturity across contracts with at least 1,700 lots of open interest. USD/CNY from FRED (DEXCHUS); SOFR (fed funds before April 2018) plus 150 bp for financing; import VAT 17%, 16% from 1 May 2018 and 13% from 1 April 2019, duty zero. Port charges 30 RMB per wet tonne and moisture 8% (per dry = per wet / 0.92). Rebar and hot-rolled coil from the Shanghai Futures Exchange; coke from Dalian.

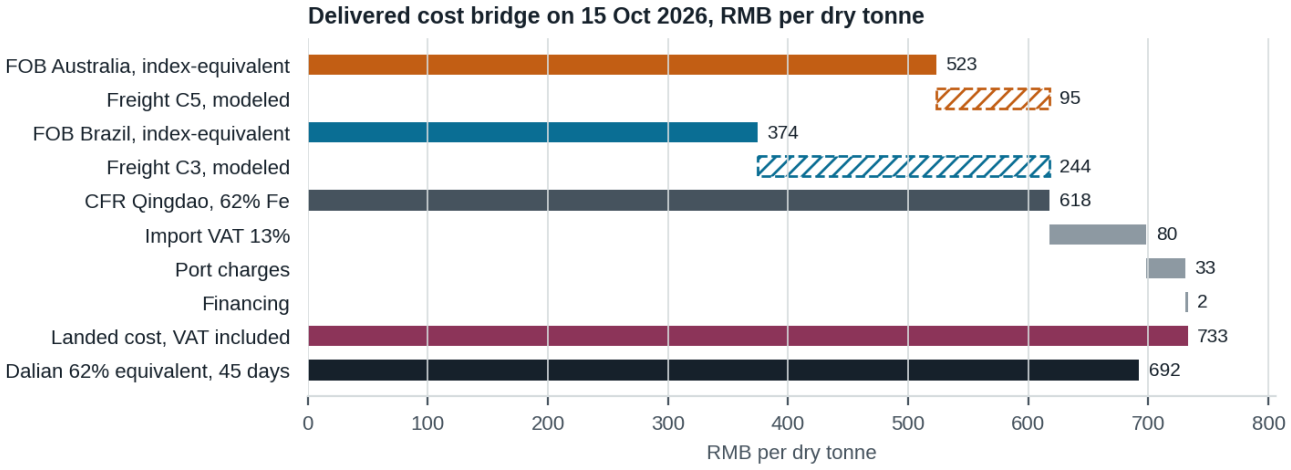


Figure 1. Freight is modeled (hatched). FOB values are index-equivalent netbacks (CFR minus freight per dry tonne) and ignore brand premiums and discounts. Sources: SGX, Dalian Commodity Exchange via Sina, FRED; freight modeled from the Baltic Capesize Index and Brent.

The import window

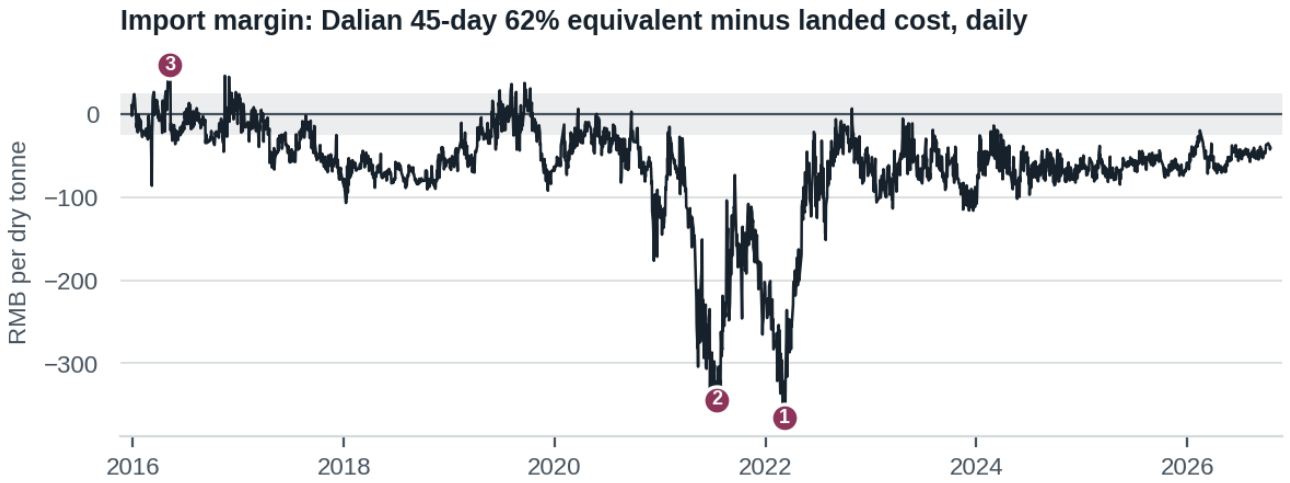


Figure 2. Shaded: the no-arbitrage band of ±25 RMB/dmt (delivery fee, about a month of warehouse storage and the vessel outbound fee cap). Numbered: the three largest deviations from the mean. Sources: SGX, Dalian Commodity Exchange, FRED; computed by the pipeline.

	Monthly baseline	Daily series
Period	Jan 2014 to Jul 2026	28 Dec 2015 to 15 Oct 2026
Observations	151	2,602
Mean, median (RMB/dmt)	−79.8, −69.4	−66.7, −57.1
Share open, inside, closed	0.0%, 3.3%, 96.7%	0.9%, 19.0%, 80.1%
Minimum	−408.9 (Jul 2021)	−366.2 (7 Mar 2022)
AR(1) half-life (95% interval)	5.0 months (2.3 to 8.2)	22.6 days (16.1 to 29.8)
ADF statistic, p-value	−2.33, 0.164	−3.42, 0.010

Table 1. Import margin statistics. Open: above +25 RMB/dmt; closed: below −25.

The three largest episodes have documented causes. (1) 7 Mar 2022, −366 RMB/dmt: Chinese regulators warned against iron ore speculation and port hoarding in February 2022, weighing on Dalian, while seaborne prices rose on supply losses in Australia and Brazil and the war in Ukraine. [NDRC price department, February 2022; NDRC and SAMR meeting with port operators, February 2022; World Bank, Commodity Markets Outlook, April 2022, p. 53] (2) 15 Jul 2021, −344 RMB/dmt: China ordered 2021 crude steel output to fall year on year. Dalian priced the cuts while seaborne 62% stayed above 200 USD/dmt; the World Bank attributes the 2021Q3 fall in iron ore prices to them. [NDRC and MIIT, 1 April 2021; World Bank, Commodity Markets Outlook, October 2021, p. 9] (3) 9 May 2016, +59 RMB/dmt: Dalian traded above import parity from 3 to 10 May 2016 after a futures rally that led the exchange to raise iron ore fees twice (to 0.9, then 1.8 per 10,000 of turnover) and the minimum margin to 8%. The reading leans on the expiring May contract. [DCE notice, 21 April 2016 (fees); DCE notice, 25 April 2016 (fees); DCE notice, 21 April 2016 (margins)]

Sensitivity of the latest monthly margin (Jul 2026, 77.4 RMB/dmt below the open threshold): Port charges (20 / 40 RMB/wmt) +10.9 / −10.9; No-arbitrage band (15 / 35 RMB/dmt) +10.0 / −10.0; Maturity target (30 / 90 days) +0.2 / −2.9; Route financing (Brazil, C3) −2.6 / −2.6; FX source (PBoC central parity) −2.4 / −2.4. No single assumption in its range opens the window. Against the most-active contract (about 108 days to expiry on average) the mean monthly margin would be −104.8 instead of −79.8 RMB/dmt, because the Dalian curve is usually backwardated.

Freight and origins

Capesize route rates (C3 Tubarao to Qingdao, C5 West Australia to Qingdao) are paid data, so freight is modeled. A voyage model converts the Baltic Capesize Index into a time charter rate (calibrated by regime on 62 dated public pairs) and Brent into a bunker price (calibrated separately before and after the IMO 2020 sulphur cap), then prices the round voyage at the Baltic eco speed of 12 knots over searoute distances (3,531 and 11,137 nautical miles). It is never fitted to a route rate; against published assessments it scores RMSE 1.04 USD/t on C5 (n = 91) and 2.04 on C3 (n = 90); a regression on the BCI and Brent scores 0.92 and 1.65 out of sample. The C3 minus C5 spread rises by about 3.3 USD/t per 1,000 BCI points (correlation 0.91). In 2026 freight takes 12.6% of CFR from Australia and 31.7% from Brazil, the highest annual shares since 2014.

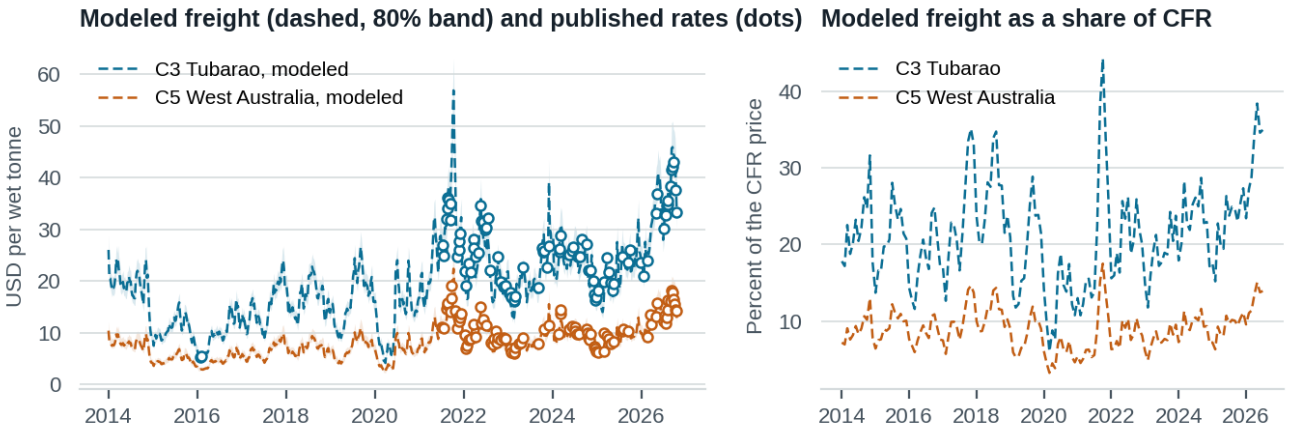


Figure 3. Left: modeled C3 and C5 with 80% error bands (dashed) and the published assessments used to score them (circles: Baltic Exchange via the shipping press, Platts, Ruida Futures). Right: modeled freight per dry tonne as a share of the 62% CFR price. Sources: East Money (BCI), FRED, IMF.

Mill margins

The Chinese market convention values a tonne of steel against 1.6 t of iron ore, 0.5 t of coke and a fixed 1,080 RMB/t of other costs (rebar; HRC 150 more), a broker convention rather than an accounting margin. The pre-registered test (written before computing it) asks whether a rebar margin in the bottom decile of its previous three years precedes weaker Dalian ore over one to three months.



Figure 4. Most-active contracts. Sources: Shanghai Futures Exchange, Dalian Commodity Exchange, Sina.

Margin	Horizon	Signal days	Effect, pp	95% interval	Newey-West t
Rebar	21 days	577	+0.97	−1.8 to +4.2	0.67
Rebar	42 days	556	+4.02	−1.0 to +9.7	1.50
Rebar	63 days	540	+5.91	−1.0 to +13.1	1.57
HRC	21 days	357	+1.52	−1.2 to +4.6	1.06
HRC	42 days	340	+5.23	+0.4 to +10.5	2.21
HRC	63 days	340	+5.31	−1.0 to +12.7	1.53

Table 2. Difference in the forward log change of the Dalian 45-day price between signal days and other days; 95% intervals from a moving block bootstrap (63-day blocks). The hypothesis was a negative effect; every estimate is positive and only HRC at 42 days has an interval that excludes zero.

Validation

Check	n	Result	Tolerance
Import cost, Ruida Futures estimates	8	max error 6.3 RMB/t with the PBoC central parity (22.2 with the New York rate)	15 RMB/t
Fastmarkets Qingdao port price to USD/dmt	5	max error 1.51 USD/dmt	2 USD/dmt
Rebar margin, broker reports	6	max error 4.4 RMB/t	5 RMB/t
C3 freight, voyage model	90	RMSE 2.04 USD/t	3.0 USD/t
C5 freight, voyage model	91	RMSE 1.04 USD/t	1.5 USD/t

Table 3. Tolerances were fixed before reconciling and are enforced by the test suite; no parameter was tuned to meet them. Each anchor, with its link, is listed on the Method and data page.

Limitations

The margin is an expected margin, not a riskless arbitrage: price, basis, grade and currency risk remain while the cargo is at sea. Dalian prices the cheapest deliverable brand with fixed quality discounts, so part of the persistent discount is a brand and grade effect. Freight is modeled, with the errors shown above, and the published anchors cluster in 2021 to 2026. Port charges are a single assumption. The daily series depends on SGX futures as a proxy for the spot index, lifted from a 61% to a 62% basis in 2026. Monthly China import volumes could not be retrieved, so the optional test of imports against the window was not run.

Sources

IMF Primary Commodity Prices via FRED (PIORECRUSDM, DEXCHUS, SOFR, DFF, DCOILBRENTEU); World Bank Pink Sheet; Dalian Commodity Exchange annual files and rules; Shanghai Futures Exchange daily files; Sina futures history; SGX daily settlement files; Baltic Capesize Index via East Money; SAFE central parity; searoute; Natural Earth. Validation: Ruida Futures daily reports, Fastmarkets, Everbright Futures, SDIC Essence Futures, Baltic Exchange route reports in the shipping press. Episode sources: NDRC, MIIT, World Bank Commodity Markets Outlook, DCE notices. Full list with links and retrieval dates: https://nathancouturier.github.io/iron-ore-china-arbitrage/#method. Code: https://github.com/nathancouturier/iron-ore-china-arbitrage.